

(12) Method to find whether a string belongs to a Grammar or not 5.6

String belongs to a Grammar or not

1) Start with the Start symbol and choose the closest production that matches to the given string

2) Replace the Variables with its most appropriate production. Repeat the process until the string is generated or until no other productions are left.

Example:

eg1 Verify $S \rightarrow 0B|A$, $A \rightarrow 0|0S|AA|G$, $B \rightarrow 1|1S|0BB$

generates the string 00110101

$S \rightarrow 0B$ ($S \rightarrow 0B$)

$\rightarrow 00BB$ ($B \rightarrow 0BB$) $\rightarrow 0011010B$ ($S \rightarrow 0B$)

$\rightarrow 0011B$ ($B \rightarrow 1$) $\rightarrow 00110101$ ($B \rightarrow 1$)

$\rightarrow 0011S$ ($B \rightarrow 1S$)

$\rightarrow 00110B$ ($S \rightarrow 0B$)

$\rightarrow 001101S$ ($B \rightarrow 1S$)



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eg.2 $S \rightarrow aAb$, $A \rightarrow aAb | \epsilon$

aabb

$S \rightarrow aAb$

$\rightarrow \underline{aa}Ab$ ($A \rightarrow aAb$)

$\rightarrow aaAb$ X or $\rightarrow aaaaAbbb$ X

$\rightarrow aaabbb$ ($A \rightarrow \epsilon$) X

(73) Derivation Tree

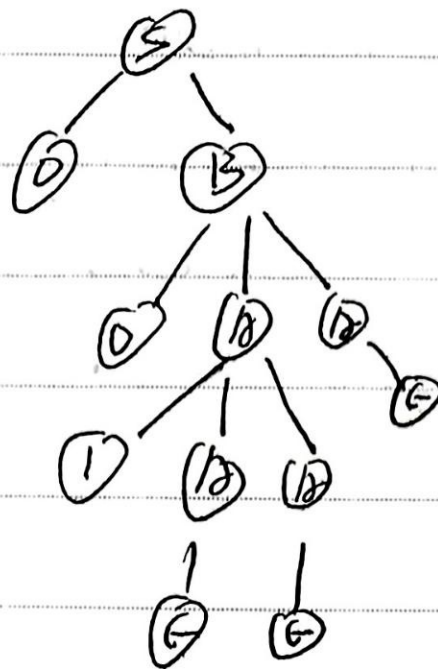
A derivation Tree or Parse Tree is an ordered *rooted tree that graphically represents the semantic information of strings derived from a Context Free Grammar

eg. $G = \{V, T, P, S\}$, where $S \rightarrow 0B$, $A \rightarrow 1AA | \epsilon$
 $B \rightarrow 0AA$

Root Vertex: Must be labelled by the Start Symbol

Vertex: Labelled by Non-Terminal Symbols

Leaves: Labelled by Terminal Symbols or ϵ



Left Derivation Tree

Right Derivation Tree

E.g. for generating the string aabaa from the Grammar $S \rightarrow aAS$
 aSS/ϵ , $A \rightarrow SBA/ba$

